

PEEQ Logic Implemented in the Crystal Plasticity UMAT

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Purpose

The implemented state variable **SDV167** stores an accumulated equivalent plastic strain, usually called PEEQ. In this UMAT, the increment is computed from the symmetric part of the plastic velocity gradient **Lp**, integrated over the current time increment **dt1**. The result is accumulated as **statev(167)**.

Implemented Equations

The plastic strain increment tensor is first formed from the symmetric part of the plastic velocity gradient:

$$\Delta \boldsymbol{\varepsilon}^p = \Delta t \text{sym}(\mathbf{L}^p) = \frac{\Delta t}{2} (\mathbf{L}^p + \mathbf{L}^{pT}). \quad (1)$$

In the code this is:

```
intLp=0.5*dt1*(Lp+transpose(Lp))
```

The tensor is then rotated into the crystal coordinate system using **Qm**:

$$\Delta \boldsymbol{\varepsilon}_c^p = \mathbf{Q}_m \Delta \boldsymbol{\varepsilon}^p \mathbf{Q}_m^T. \quad (2)$$

In the code this is:

```
intLp=matmul(Qm, matmul(intLp, transpose(Qm)))
```

The six independent components of this plastic strain increment are accumulated in **SDV156** through **SDV161**:

$$\text{SDV156} \leftarrow \text{SDV156} + \Delta \varepsilon_{c,11}^p, \quad (3)$$

$$\text{SDV157} \leftarrow \text{SDV157} + \Delta \varepsilon_{c,22}^p, \quad (4)$$

$$\text{SDV158} \leftarrow \text{SDV158} + \Delta \varepsilon_{c,33}^p, \quad (5)$$

$$\text{SDV159} \leftarrow \text{SDV159} + \Delta \varepsilon_{c,12}^p, \quad (6)$$

$$\text{SDV160} \leftarrow \text{SDV160} + \Delta \varepsilon_{c,13}^p, \quad (7)$$

$$\text{SDV161} \leftarrow \text{SDV161} + \Delta \varepsilon_{c,23}^p. \quad (8)$$

For PEEQ, only the deviatoric part is used. The code stores one third of the trace in **trLp**:

$$m = \frac{1}{3} (\Delta \varepsilon_{c,11}^p + \Delta \varepsilon_{c,22}^p + \Delta \varepsilon_{c,33}^p). \quad (9)$$

The deviatoric increment is then:

$$\Delta \mathbf{e}_c^p = \Delta \boldsymbol{\varepsilon}_c^p - m \mathbf{I}. \quad (10)$$

The equivalent plastic strain increment is the standard J2 equivalent measure:

$$\Delta \bar{\varepsilon}^p = \sqrt{\frac{2}{3} \Delta \mathbf{e}_c^p : \Delta \mathbf{e}_c^p}. \quad (11)$$

For a symmetric tensor, the double contraction used by the code is:

$$\Delta \mathbf{e}_c^p : \Delta \mathbf{e}_c^p = e_{11}^2 + e_{22}^2 + e_{33}^2 + 2(e_{12}^2 + e_{13}^2 + e_{23}^2). \quad (12)$$

Therefore the code expression is:

```
peeq_inc=sqrt(2.0d0/3.0d0*
&          (devLp(1,1)**2+devLp(2,2)**2+devLp(3,3)**2
&          +2.0d0*(devLp(1,2)**2+devLp(1,3)**2
&          +devLp(2,3)**2)))
```

Finally, the accumulated PEEQ is stored in SDV167:

$$\text{SDV167}_{n+1} = \text{SDV167}_n + \Delta \bar{\varepsilon}^p. \quad (13)$$

In the code this is:

```
statev(167)=statev(167)+peeq_inc
```

Initialization and Abaqus Requirement

At the beginning of the analysis, the value is initialized as:

```
statev(167)=0
```

The Abaqus input file must declare enough solution-dependent state variables. Because this implementation writes to **SDV167**, the input deck must use ***Depvar** with at least 167 state variables. The example input decks in the repository use 200, which is sufficient.

Interpretation

This implementation gives an accumulated scalar plastic strain measure based on the deviatoric part of the plastic strain increment. Since **Qm** is a rotation matrix, the J2 norm is invariant under the coordinate transformation; using the rotated tensor is also consistent with the existing storage of **SDV156–SDV161** in crystal coordinates.